Semester One Examination, 2025

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section Two: Calculator-assumed

If required by your examination administrator, please place your student identification label in this box

WA student number: In figures

In words

Circle your Teacher's Name: Mrs Bestall Mr Buckland Mrs Ducat Mrs Fraser-Jones Mr Kim Mr Lei Mr Luzuk Ms Narendranathan Mr Ng Mr Safa Mr Tanday

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Number of additional answer booklets used (if applicable):

Materials required/recommended for this section

To be provided by the supervisor
This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet preferably using a blue/black pen.
Do not use erasable or gel pens.

3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.

4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

5. It is recommended that you do not use pencil, except in diagrams.

6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
8	7	
9	7	
10	9	
11	9	
12	9	
13	7	
14	8	
15	9	
16	8	
17	8	
18	7	
19	10	
Section 2 Total	98	
Weighted (×0.6633)	65%	

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Section Two: Calculator-assumed **65% (98 Marks)**

This section has **twelve** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 8 **(7 marks)**

The universal set U contains the 26 uppercase letters of the alphabet and sets X, Y and Z are

$$X = \{P, R, O, F, U, S, E\}, \qquad Y = \{C, H, A, L, K, Y\}, \qquad Z = \{P, N, E, U, M, O, G, A, S, T, R, I, C\}$$

(a) Determine the set $X \cap Z$. (1 mark)

(b) Use set notation to describe the set of letters that are in set Y or not in set Z. (1 mark)

(c) State, with justification, which two sets are mutually exclusive. (2 marks)

(d) Determine

(i) $n(\overline{X \cap Z})$. (1 mark)

(ii) $P(X \cup Y \cup Z)$. (2 mark)

Question 9 **(7 marks)**

(a) State the equation of the line of symmetry of the graph of $y = 4x^2 - 32x + 51$. (1 mark)

(b) State the roots of the graph of $y = -2(x+1)(x-3)(x-8)$. (1 mark)

(c) Let $f(x) = 2\sqrt{x+4}$ for $0 \leq x < 32$. Determine the range of $f(x)$. (2 marks)

(d) The function $g(x) = ax^2 + bx + c$ has a turning point at $(5, -8)$ and passes through the point $(2, 10)$. Determine the value of each of the constants a, b and c. (3 marks)

Question 10 **(9 marks)**

Isaiah creates a biased die with uneven odds to use to win at games. The frequency of each integer generated during a testing session was recorded in the table below and then used to calculate the relative frequency.

Integer	1	2	3	4	5	6
Frequency	30	28	15	42		41
Relative frequency	0.15	0.14	0.075	0.21		0.205

(a) Show that 200 rolls were conducted during the testing session. (1 mark)

(b) Complete the two missing entries in the table above. (2 marks)

(c) Use the testing session data to determine the probability that

(i) the next roll will be greater than 2. (2 marks)

(ii) the next integer generated will be 3, given that it is less than 4. (2 marks)

(iii) the next two integers generated will not both be 2. (2 marks)

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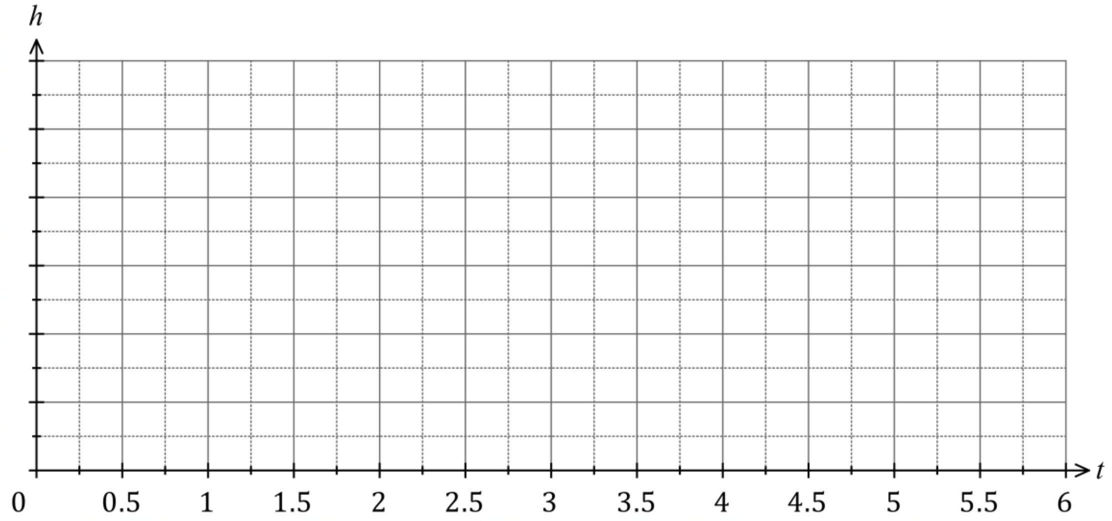
Question 11 **(9 marks)**

Jiyeong goes bungee jumping and moves in a vertical line so that at time t seconds, her height h metres above the ground is given by

$$h = 60 + 54\cos\left(\frac{2\pi t}{3}\right)$$

(a) By first adding a suitable scale to the vertical axis, sketch the graph of h against t for $0 \leq t \leq 6$. (4 marks)



The closest that Jiyeong ever comes to the ground is d metres.

(b) State the value of d. (1 mark)

(c) Determine the number of times that Jiyeong is d metres above the ground during the first minute. (2 marks)

(d) Determine the total time for which Jiyeong is at least 87 metres above the ground between $t = 0$ and $t = 4$. (2 marks)

Question 12 **(9 marks)**

(a) Evaluate the number of combinations given by $\binom{33}{3}$. (1 mark)

(b) State the value(s) of n and the value(s) of r when $\binom{n}{r} = \frac{n!}{8!\,5!}$. (2 marks)

Sophia can only fit 3 desserts on a single plate at a buffet. So, Sophia selects 3 desserts from 13 desserts offered at the buffet, 7 of which are hot and the remainder cold.

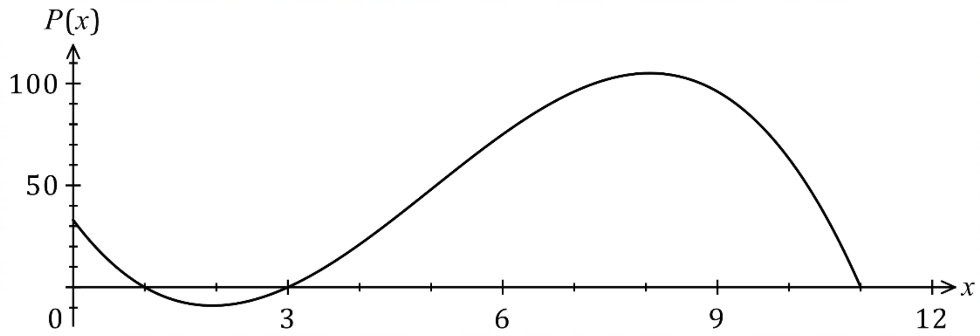
(c) How many different selections are possible? (2 marks)

(d) How many different selections are possible if they must all be cold? (2 marks)

(e) If the selection is made at random, determine the probability that all the desserts chosen are hot. (2 marks)

Question 13 **(7 marks)**

The function defined by $P(x) = 33 - 47x + 15x^2 - x^3$ for $0 \le x \le 11$ was used to model the monthly profit P, in thousands of dollars, when a factory set the production rate of a chemical to x thousand litres per day. The graph of $y = P(x)$ is shown.



(a) Determine $P(2.7)$ and interpret your answer in the context of the question. (2 marks)

(b) Express $P(x)$ in factored form. (1 mark)

(c) State the range of the function for the given domain of $0 \le x \le 11$. (1 mark)

(d) By only considering production rates in the interval $0 \le x \le 8$, determine the change in the production rate required to increase the monthly profit by $\$30\ 000$ from its value when the factory is producing 4400 litres per day. (3 marks)

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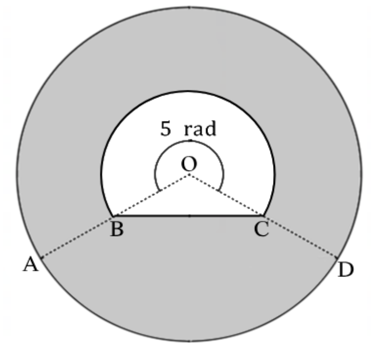
Question 14 **(8 marks)**

The diagram shows a circle with a hole through the middle.



$\angle AOD = 5$ radians and
$AB = BO = OC = CD = 3.3$ cm

(a) Determine the perimeter of the shaded region to 2d.p. (4 marks)

(b) Determine the area of the shaded region to 2d.p. (4 marks)

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Question 15 **(9 marks)**

A group of Rossmoyne students are asked whether they study Economics or History. Analysis of the 399 responses showed that 57 study both, 190 study neither and 171 study Economics.

(a) Determine how many of these students study History but not Economics. (2 marks)

(b) One of the students is chosen at random. Determine the probability that they

(i) study Economics or History. (2 marks)

(ii) study History given that they study Economics. (2 marks)

(c) For these students, are the events studying Economics and studying History independent? Justify your answer. (3 marks)

Question 16 **(8 marks)**

A small tin and a large tin each contain a mixture of green and red plastic counters. There are 7 green and 4 red counters in the small tin, and 6 green and 8 red counters in the large tin.

(a) When two counters are removed at random and without replacement from the large tin, determine the probability that they are both green. (2 marks)

(b) When one counter is removed at random from each tin, determine the probability that the counters are the same colour. (3 marks)

(c) When a counter is removed at random from the large tin and placed in the small tin, determine the probability that the next counter drawn at random from the small tin is red. (3 marks)

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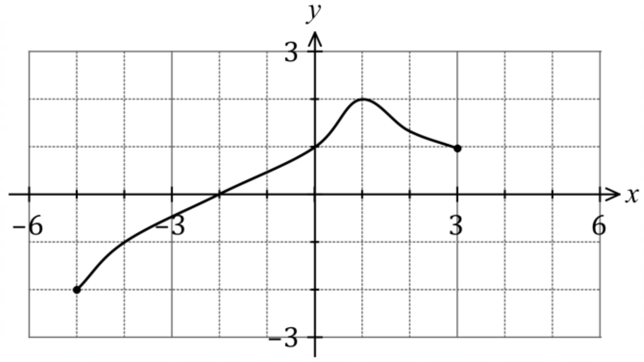
Question 17 **(8 marks)**

The graph of $y = f(x)$ is shown.



(a) Determine $f(-4)$. (1 mark)

(b) State the domain of the function f. (1 mark)

(c) State the range of the function f. (1 mark)

(d) Solve $f(x) = 1$. (1 mark)

(e) Determine the **coordinates** of the following points:

(i) the x-axis intercept of the graph of $y = f(x - 3)$. (1 mark)

(ii) the y-axis intercept of the graph of $y = -8f(x)$. (1 mark)

(iii) the turning point of the graph of $y = f(0.5x) - 3$. (2 marks)

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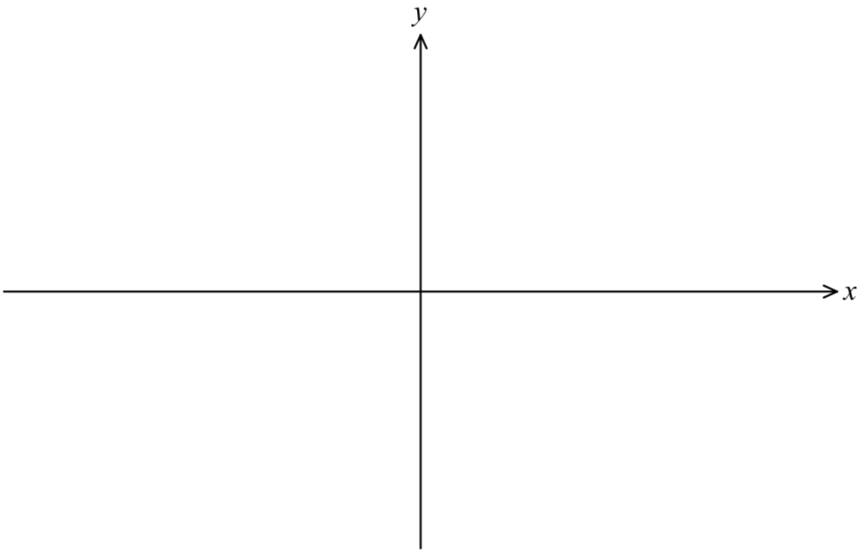
Question 18 **(7 marks)**

The graph of $y = f(x)$, where f is a cubic polynomial, has a turning point at $\left(-{}^{5}/_{2}, 0\right)$ and passes through the points with coordinates $(0, -200)$ and $\left({}^{4}/_{3}, 0\right)$.

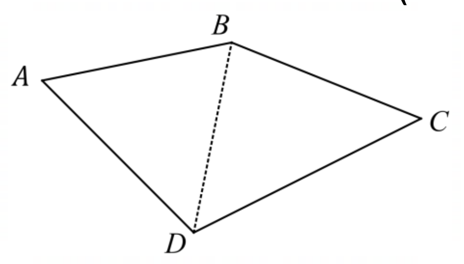
(a) Sketch the graph of $y = f(x)$ on the axes below. (3 marks)



(b) Determine $f(x)$ in expanded form $ax^3 + bx^2 + cx + d$ where a, b, c and d are constants. (4 marks)

Question 19 **(10 marks)**

Ann buys a plot of land to start farming. The diagram shows a sketch of the plot of land $ABCD$ with diagonal BD.



Known lengths (in metres) are $CD = 62.5$, $BC = 46.5$ and $BD = 48.3$, and known angles are $\angle ADB = 54°$ and $\angle ABD = 56.5°$.

(a) Show use of trigonometry to determine the size of $\angle BCD$. (2 marks)

(b) Show use of trigonometry to determine the length of the side AD. (2 marks)

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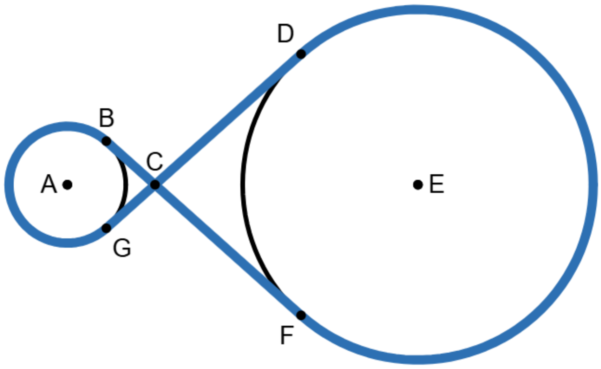
(c) On Ann's farm, there are two water towers that are 1m and 3m in radius, and 2m apart. Ann takes a can of spray paint and draws on the ground, around the two water tanks and connects the paint together as shown in the diagram. The intersection of the paint trail at C is 1.5m away from the centre of the smaller water tower.
What is the length of this paint trail? (6 marks)
Note: Use radian measure



End of questions

Supplementary page

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